

ReFit: Repairing Speculative Drafters for Post-Trained Language Models

Anonymous ACL submission

Abstract

Speculative decoding with target-conditioned drafters such as EAGLE-3 is among the fastest ways to serve large language models, and the drafter released with a base model is routinely reused for its many post-trained derivatives. We show that this reuse breaks for an important and growing class of models: derivatives distilled from a stronger reasoning teacher. In a census of 174 public derivatives, quantized, merged and RL-tuned models retain 98–100% of the family drafter’s acceptance, while DeepSeek-R1-Distill-Llama-8B and Llama-3.1-Nemotron-Nano-8B lose up to 30%, erasing most of the speedup. Post-training changes both the hidden states the drafter reads and the tokens it must predict, and the most effective place to repair the drafter is its *feature interface*, the projection through which it reads the target. We introduce *ReFit*, which re-fits a family drafter on the derivative’s own responses to public instructions, without any of the original training data. Re-fitting the interface alone (12% of the drafter’s parameters) closes 60% of the gap to a dedicated drafter trained specifically for the derivative, and a full warm start closes 72%, using 11 GPU-hours, over $400\times$ less compute than the public recipe for a dedicated drafter. On SPEED-Bench and MATH-500, ReFit raises the end-to-end speedup of R1-Distill from $1.31\times$ to $1.83\times$ (dedicated drafter: $2.05\times$) and of Nemotron-Nano from $1.29\times$ to $1.80\times$, outperforming independent draft models, and its gains extend to sampled decoding and to reasoning traces thousands of tokens long.

1 Introduction

Speculative decoding has become the standard way to serve large language models (LLMs) quickly: a small drafter proposes several tokens, and the target model verifies all of them in a single forward pass, accelerating generation without changing its output (Leviathan et al., 2023; Chen et al., 2023).

The fastest drafters today are *target-conditioned*. Rather than modeling text from scratch, they read the target’s own hidden states and predict its next tokens from them (Cai et al., 2024; Li et al., 2024b, 2025; Chen et al., 2026b). EAGLE-3, for instance, fuses features from the target’s lower, middle and upper layers and reports speedups of up to $6.5\times$ (Li et al., 2025). This tight coupling is also their price: each drafter is trained for one specific target, on hundreds of thousands of examples, for tens of epochs.

Open-weight models, however, rarely stay single models. A model family quickly acquires instruction-tuned releases, quantized and merged variants, preference-optimized and RL-tuned checkpoints, and reasoning distillations; a popular base model such as Llama-3.1-8B spawns thousands of such derivatives on public model hubs (Horwitz et al., 2025). Training a dedicated drafter for each one is out of reach, so serving stacks reuse the drafter released for the family’s instruction model, the *family drafter*, for all of its relatives (Kwon et al., 2023; Red Hat AI and vLLM Project, 2025). Practitioners make this choice every day, yet two questions behind it remain open: when does a family drafter keep working for a derivative, and what can be done when it does not?

We answer the first question with a census of 174 public derivatives in the Llama-3.1-8B and Qwen3-8B families (Figure 1a). For most of the ecosystem, reuse is reliable. Quantized, merged, ablated and on-policy RL derivatives retain 98–100% of the family drafter’s acceptance, in line with evidence that on-policy training keeps models close to their initialization (Shenfeld et al., 2026; Chen et al., 2026a). Reuse fails, however, for precisely the derivatives that are most expensive to serve: models distilled from stronger reasoning teachers. DeepSeek-R1-Distill-Llama-8B and Llama-3.1-Nemotron-Nano-8B lose 26–30% of first-token acceptance under both EAGLE-3

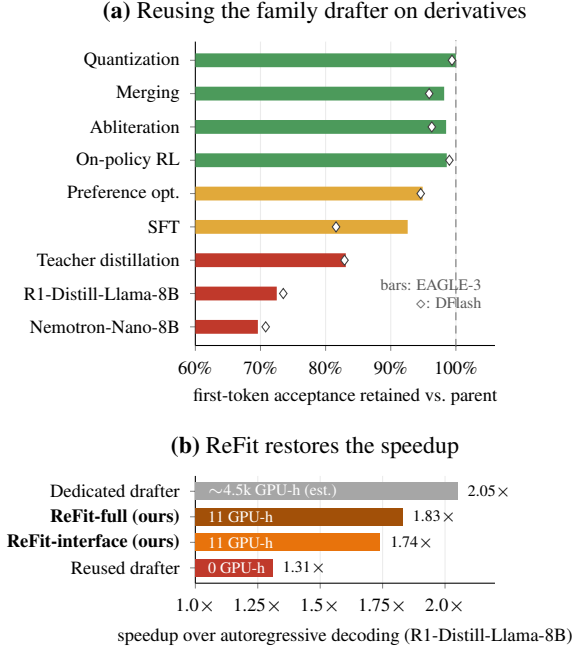


Figure 1: **(a)** First-token acceptance of the Llama-3.1-8B-Instruct family drafters on public derivatives, relative to the instruction model they were trained for (174 derivatives grouped by post-training history; EAGLE-3 bars, DFlash diamonds). Weight-space edits and on-policy RL leave the drafter intact; distillation from a reasoning teacher breaks it. **(b)** On R1-Distill-Llama-8B, ReFit recovers most of the speedup of a dedicated drafter at a small fraction of its training cost.

and DFlash drafters. For R1-Distill, the speedup collapses from $2.05\times$ with a dedicated drafter to $1.31\times$ (Figure 1b). Reasoning models routinely emit thousands of tokens per query, so this lost speedup costs the most exactly where acceleration matters most.

We then show where the damage lies. Post-training changes two things the drafter depends on: the target’s hidden states, which the drafter reads, and the target’s next-token policy, which it must predict. A controlled crossover shows that both shifts are substantial. The drafter, however, accesses the target through a single learned map, the *feature interface*, a dense projection that compresses the target’s multi-layer hidden states into the drafter’s input space. With the same training recipe and parameter budget, re-fitting this interface recovers more than updating the drafter’s decoder, and the interface alone recovers most of what re-training the entire drafter achieves. This observation leads to *ReFit* (Figure 3): elicit the derivative’s own responses to public instructions,

then re-fit the family drafter on them with its native training objective, updating either the interface alone (ReFit-interface) or the whole warm-started drafter (ReFit-full). ReFit needs no access to the derivative’s or the drafter’s training data, no new architecture, and no change to the serving engine.

ReFit is highly effective. On R1-Distill-Llama-8B, ReFit-interface closes 60% of the gap between the reused family drafter and a dedicated drafter trained by the EAGLE authors, and ReFit-full closes 72%. With 11 GPU-hours of compute, over $400\times$ less than the public recipe for a dedicated drafter, ReFit lifts the end-to-end speedup from $1.31\times$ to $1.83\times$. On Nemotron-Nano, for which no dedicated drafter exists, it lifts the speedup from $1.29\times$ to $1.80\times$. Repaired drafters are faster than independent small draft models and training-free drafting, and the gains hold across drafter checkpoints, drafter families, model families, long reasoning traces and sampled decoding. In summary, we make three contributions:

- We present a census of family-drafter reuse across 174 public derivatives and identify off-policy distillation as the post-training regime that breaks target-conditioned drafters (§3).
- We trace the failure to shifted target features and policy, identify the drafter’s feature interface as the most effective place to repair it, and propose ReFit, a simple recipe that repairs the family drafter from the derivative’s own outputs without any original training data (§3–§4).
- We show on SPEED-Bench and MATH-500 that ReFit recovers most of a dedicated drafter’s speedup at a small fraction of its cost, across targets, drafter checkpoints and drafter families (§6).

2 Background: Target-Conditioned Drafters

Speculative decoding. Given a prefix $y_{\leq t}$, a drafter proposes K tokens $\hat{y}_{t+1:t+K}$. The target scores all of them in one forward pass and accepts the longest prefix that agrees with its own predictions, plus one bonus token that it generates itself (Leviathan et al., 2023; Chen et al., 2023). The output is identical to decoding with the target alone. Efficiency is governed by the acceptance length τ , the average number of tokens produced per target forward pass, and by the cost of drafting. With per-

step costs c_T for the target and c_D for the drafter, the speedup is approximately $\tau c_T / (c_T + K c_D)$. A good drafter is therefore both *accurate* (high τ) and *cheap* (low c_D). We also report first-token acceptance p_1 , the rate at which the target accepts the first drafted token; unlike τ , it does not depend on how long the target’s responses are.

Target-conditioned drafting. Target-conditioned drafters achieve both properties by reusing the target’s computation. Medusa attaches extra decoding heads to the target’s final layer (Cai et al., 2024), and EAGLE drafts autoregressively at the feature level (Li et al., 2024b,a). EAGLE-3 (Li et al., 2025) reads hidden states $h_t^{\text{low}}, h_t^{\text{mid}}, h_t^{\text{high}} \in \mathbb{R}^d$ from three depths of the target and maps them through a learned *feature interface* $W_{\text{fc}} \in \mathbb{R}^{d \times 3d}$:

$$g_t = W_{\text{fc}} [h_t^{\text{low}}; h_t^{\text{mid}}; h_t^{\text{high}}]. \quad (1)$$

The fused feature g_t is combined with the embedding of the next token and passed through a single decoder layer and a language-modeling head over a reduced draft vocabulary, producing the draft distribution q_θ . EAGLE-3 is trained with *training-time test*: the drafter is unrolled on its own predictions for $k = 1, \dots, k_{\text{max}}$ steps and matched to the target’s next-token distribution p_T at each step,

$$\mathcal{L}(\theta) = \sum_{k=1}^{k_{\text{max}}} \mathbb{E}_t \text{KL} \left(p_T(\cdot \mid y_{\leq t+k}) \parallel q_\theta^{(k)} \right), \quad (2)$$

so that it learns to draft from its own predictions as it does at inference. DFlash (Chen et al., 2026b) drafts an entire block in parallel with a lightweight diffusion model, but it reads the target through the same kind of multi-layer feature projection.

Family drafters. A drafter is trained once for a family’s instruction model and released publicly. Serving engines then load it for any model that shares that model’s architecture and tokenizer (Kwon et al., 2023; Red Hat AI and vLLM Project, 2025). We call the instruction model the *parent* T , its drafter the *family drafter* D_{θ_0} , and the model being served the *derivative* T' .

3 When Family Drafters Fail

A census of derivatives. We collect 174 public derivatives in the Llama-3.1-8B and Qwen3-8B families (Grattafiori et al., 2024; Yang et al., 2025) from the Hugging Face hub, all of which load with

the architecture and tokenizer of the instruction-tuned parent and its released drafters. We label each by its post-training history as described in its model card: quantization, merging, ablation, supervised fine-tuning (SFT), preference optimization, on-policy reinforcement learning, or distillation from a teacher. For every derivative, we serve the family EAGLE-3 and DFlash drafters with vLLM (Kwon et al., 2023) and measure first-token acceptance on identical prompts for the derivative and its parent. We report retention, the derivative’s acceptance divided by the parent’s.

Distillation breaks the drafter. Figure 1a shows a clear ordering. Edits in weight space leave the drafter intact: quantized derivatives retain 100% of the parent’s acceptance, merged ones 98% and ablated ones 99%. On-policy RL does not disturb it either (99%). Off-policy supervision erodes acceptance progressively, from 95% for preference optimization and 93% for SFT to 83% for distillation from a teacher. The most severe losses occur for the two widely used reasoning distills. R1-Distill-Llama-8B, trained on the outputs of DeepSeek-R1 (DeepSeek-AI, 2025), retains 72.5% under EAGLE-3 and 73.5% under DFlash. Nemotron-Nano-8B (Bercovich et al., 2025) retains 69.6% and 70.8%. This ordering mirrors how far each regime moves a model from its initialization. On-policy RL learns from the model’s own samples and stays close to it (Shenfeld et al., 2026; Chen et al., 2026a), whereas distillation fits the model to a different model’s distribution (Hinton et al., 2015; Kim and Rush, 2016) and reshapes its internal representations.

The cost of this loss is substantial. On R1-Distill, the reused drafter reaches an acceptance length of only 1.76 tokens per step on SPEED-Bench, compared with 2.85 for a dedicated EAGLE-3 drafter that the EAGLE authors trained on R1-Distill. In end-to-end decoding, this is the difference between $1.31 \times$ and $2.05 \times$ (§6). A dedicated drafter is rarely an option: the public EAGLE-3 recipe trains on over half a million examples for tens of epochs, and most derivatives, including Nemotron-Nano, have none. Practitioners can, however, tell cheaply when reuse is at risk: acceptance measured on just 16 requests ranks the degraded derivatives in our populations with an AUROC between 0.91 and 1.00.

Both the features and the policy move. A drafter can fail on a derivative for two reasons: the

derivative presents it with different hidden states, or the derivative wants different next tokens. We separate the two with a fixed-text crossover. On 64 contexts generated by each derivative, we feed the family drafter either the parent’s or the derivative’s hidden states and score its first prediction against either model’s next token. Swapping in the derivative’s hidden states alone lowers agreement by 2.0 points on R1-Distill (95% CI $[-2.5, -1.4]$) and by 6.9 points on Nemotron-Nano ($[-7.7, -6.2]$). Swapping in the derivative’s next-token policy lowers it by another 3.0 ($[-3.5, -2.6]$) and 3.7 points ($[-4.3, -3.1]$). The family drafter thus misreads the derivative’s features even when the desired token is unchanged, and it also aims at the wrong tokens. An effective repair must teach the drafter to read the derivative’s features, under supervision from the derivative itself.

The interface is the place to repair. Which part of the drafter should learn? We re-fit different parts of the R1-Distill family drafter on 256 of the derivative’s responses with one training recipe (§4) and measure how much of the gap to the dedicated drafter each closes (Figure 2). At every parameter budget we compare, the interface is the better place to spend it. With 1.2M trainable parameters, a low-rank update of the interface closes 22% of the gap, while a decoder update of identical size closes 10%. With 50M parameters, the dense interface closes 36%, more than a full-rank update of the decoder’s attention projections or a decoder adapter of the same size (33% each). Recalibrating the mean and scale of the target’s features closes none of the gap: the derivative’s features call for a learned re-mapping, not a shift and rescaling. Larger updates recover more, with the whole decoder closing 43% and the entire drafter 46%, but the interface alone captures most of this gain with an eighth of the parameters. These two points, the dense interface and the entire warm-started drafter, are the two operating points of ReFit.

4 ReFit: Repairing the Family Drafter

Given a derivative $\mathcal{T}'$ and the family drafter D_{θ_0} of its parent, ReFit produces a drafter D_{θ} matched to $\mathcal{T}'$ without the data used to train either model. The recipe has three steps (Figure 3).

Eliciting the derivative. The drafter must predict what $\mathcal{T}'$ will generate, so we let $\mathcal{T}'$ show it. We sample instructions x from a public instruc-

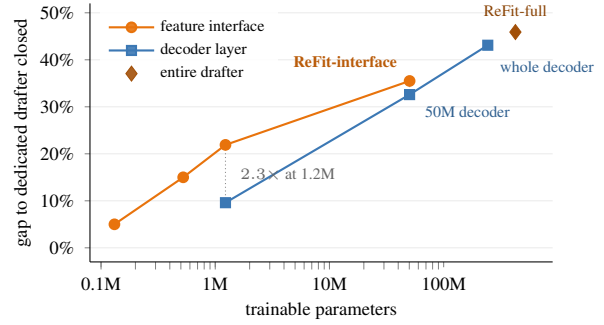


Figure 2: **Where to repair.** Fraction of the acceptance-length gap between the reused family drafter and a dedicated drafter closed by re-fitting the feature interface or the decoder layer with a given number of trainable parameters (R1-Distill-Llama-8B, SPEED-Bench; same 256 repair examples and training recipe for every point). At equal capacity the interface recovers more, and the dense interface (ReFit-interface) captures most of what re-training the entire drafter (ReFit-full) achieves.

tion corpus. We use 16k instructions from Alpaca (Taori et al., 2023), deduplicated against all evaluation prompts. We render each instruction with the derivative’s chat template and greedily decode a response $y' \sim \mathcal{T}'(\cdot | x)$ of up to 512 tokens. For a reasoning distill, these responses are long reasoning traces, exactly the text the drafter will face in deployment. This mirrors sequence-level distillation (Kim and Rush, 2016) and drafter alignment (Zhou et al., 2024; Goel et al., 2024), with one difference: only the drafter learns, and only to follow a fixed target.

Re-fitting the drafter. For each pair (x, y') , a single teacher-forced pass of $\mathcal{T}'$ provides, at every response position, the three hidden states that enter Eq. (1) and the next-token distribution $p_{\mathcal{T}'}$. These two signals address exactly the two shifts identified in §3: the drafter learns to read $\mathcal{T}'$ ’s features and to predict $\mathcal{T}'$ ’s tokens. We minimize the drafter’s native training-time-test objective (Eq. 2) with $p_{\mathcal{T}'}$ in place of p_T and $k_{\max} = 3$, starting from the family drafter’s parameters θ_0 . We consider two operating points:

- **ReFit-interface** updates only W_{fc} , 50M parameters or 12% of the drafter, and keeps the decoder layer and head frozen.
- **ReFit-full** warm-starts all drafter parameters from θ_0 ; the token embedding remains shared with $\mathcal{T}'$.

Both variants inherit everything the family drafter

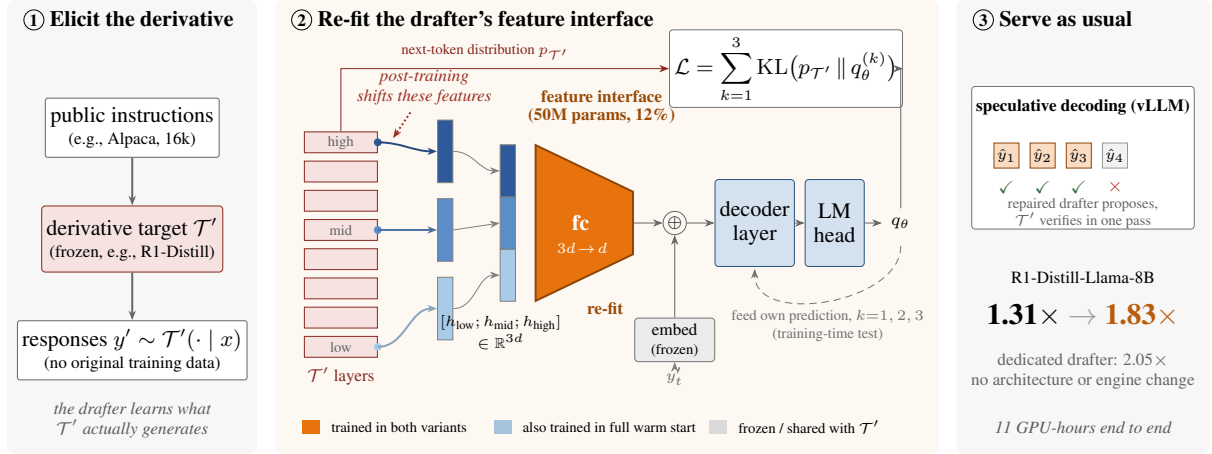


Figure 3: **ReFit**. ① The derivative target $\mathcal{T}'$ answers public instructions, producing data that reflect what it actually generates; none of its training data, nor the drafter’s, is needed. ② A teacher-forced pass of $\mathcal{T}'$ exposes the multi-layer hidden states that the drafter reads and the next-token distribution it must match. Post-training has shifted both, so we re-fit the drafter’s feature interface (orange, ReFit-interface) with its native training-time-test objective, optionally warm-starting the decoder layer and head as well (blue, ReFit-full). ③ The repaired drafter keeps its architecture and format, so it is served with an unchanged engine.

already knows about drafting and only teach it to read the new target. This warm start is essential. The same architecture trained from scratch on the same data and budget falls below even the reused drafter (§6).

Cost and deployment. ReFit takes one pass over 16k examples, or 7.9M response tokens. Generating the responses and training together take about 11 A40 GPU-hours, of which training the drafter accounts for only 2.2. The repaired drafter has the same architecture, vocabulary and file format as the family drafter, so it is a drop-in replacement in the serving engine.

5 Experimental Setup

Targets and drafters. We study the two reasoning distills from the census: DeepSeek-R1-Distill-Llama-8B, distilled from DeepSeek-R1 into Llama-3.1-8B (DeepSeek-AI, 2025), and Llama-3.1-Nemotron-Nano-8B-v1, derived from Llama-3.1-8B-Instruct through distillation and reasoning post-training (Bercovich et al., 2025). Both share the Llama-3.1 architecture and tokenizer, and hence the family drafters of Llama-3.1-8B-Instruct. Our main family drafter is the EAGLE-3 drafter that the EAGLE authors released for Llama-3.1-8B-Instruct (Li et al., 2025). We also repair the production EAGLE-3 drafter distributed with vLLM’s speculators library (Red Hat AI and vLLM Project, 2025) and the DFlash drafter (Chen et al., 2026b). As an upper reference, we use the dedi-

cated EAGLE-3 drafter that the EAGLE authors trained specifically for R1-Distill-Llama-8B.

Baselines. We compare against (i) reusing the family drafter unchanged; (ii) an independent draft model, Llama-3.2-1B-Instruct (Grattafiori et al., 2024), which shares the targets’ tokenizer; (iii) n-gram prompt lookup (Saxena, 2023), a training-free drafter built into vLLM; and (iv) training the same EAGLE-3 architecture from scratch on our repair data with the same budget. All speculative methods propose $K=4$ tokens per step.

Benchmarks and metrics. We evaluate on SPEED-Bench (Abramovich et al., 2026), a unified benchmark for speculative decoding, using a stratified 128-prompt sample of its qualitative split, which spans eleven domains from coding and math to writing, roleplay and multilingual chat. For long-form reasoning, we use MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024). Prompts are rendered with each target’s chat template, and responses are generated greedily for up to 512 tokens. We report p_1 , τ (including the bonus token) and, for R1-Distill, the *gap closed*, $(\tau_{\text{repair}} - \tau_{\text{reuse}}) / (\tau_{\text{dedicated}} - \tau_{\text{reuse}})$. We measure end-to-end wall-clock speedup over autoregressive decoding in vLLM on NVIDIA A40 GPUs at batch sizes 1 and 8. R1-Distill results on SPEED-Bench average three training seeds; confidence intervals are 95% paired bootstrap intervals over prompts (and seeds). Further details are in Appendix A.

6 Results

6.1 Main Results

Table 1 summarizes our results. On R1-Distill-Llama-8B, ReFit-interface raises the acceptance length on SPEED-Bench from 1.76 to 2.41 (paired gain 0.650 [0.607, 0.692]) and first-token acceptance from 43.0% to 62.6%, closing 59.9% [57.6, 62.4] of the gap to the dedicated drafter while keeping the decoder fixed. ReFit-full raises the acceptance length to 2.54 (0.778 [0.731, 0.823]) and first-token acceptance to 65.0%, within 5.5 points of the dedicated drafter, closing 71.7% [69.2, 74.3] of the gap. The gains are equally strong on long mathematical reasoning: on MATH-500, ReFit raises τ from 1.93 to 2.77 and 2.97, closing 44% and 54% of the gap. On Nemotron-Nano, where no dedicated drafter exists and repair is the only way to obtain a matched drafter, ReFit-interface and ReFit-full raise τ by 0.655 [0.604, 0.707] and 0.724 [0.662, 0.787] on SPEED-Bench, and by 1.01 and 1.21 on MATH.

These gains carry over to wall-clock time. At batch size 1, the repaired drafters accelerate R1-Distill by 1.83 $\times$ and 1.74 $\times$, against 1.31 $\times$ for the reused drafter and 2.05 $\times$ for the dedicated one. ReFit-full thus delivers 79% of the dedicated drafter’s acceleration over autoregressive decoding for an estimated 0.25% of its training compute. On Nemotron-Nano, the speedup rises from 1.29 $\times$ to 1.80 $\times$. Measured directly against reuse on the same prompts and server processes, ReFit-full decodes 1.39 $\times$ faster on both targets (95% CI [1.32, 1.47] on R1-Distill and [1.31, 1.48] on Nemotron-Nano). The improvements persist under batching, where speculative decoding has less idle compute to exploit: at batch size 8, ReFit lifts the speedup from 1.07 $\times$ to 1.29 $\times$ on R1-Distill and from 1.13 $\times$ to 1.42 $\times$ on Nemotron-Nano.

What makes ReFit work. Two ingredients matter beyond the choice of interface. The first is the warm start. Training the same EAGLE-3 architecture from scratch on the repair data reaches an acceptance length of only 1.52 on SPEED-Bench, below even the unrepaired family drafter and a full token per step below ReFit-full. A drafter’s ability to draft is expensive to acquire, and the family drafter already has it. The second is supervision from the derivative itself. Training on the same derivative-generated text, but with the parent’s hidden states and next-token distributions as supervi-

sion, also improves the drafter because it adapts to the new text. Supervision from the derivative, however, adds a further 0.15 tokens per step for the interface and 0.19 for the entire drafter on SPEED-Bench (0.14 and 0.15 on MATH), with confidence intervals well above zero. ReFit needs both the derivative’s text and the derivative’s signals, which is exactly what eliciting the derivative provides.

ReFit versus other drafters. Training-free n-gram lookup rarely finds long matches in open-ended generation and stays below 1.6 tokens per step on both targets. The independent Llama-3.2-1B drafter predicts the derivatives well, but it runs sixteen transformer layers per drafted token, and this cost holds it to 1.26 $\times$ on R1-Distill and 1.37 $\times$ on Nemotron-Nano. Repaired drafters combine high acceptance with the near-free drafting of a single-layer, target-conditioned head and are the fastest option on both targets and at both batch sizes.

6.2 Long Reasoning and Sampling

Reasoning models are deployed with long generation budgets and often with sampling, whereas ReFit sees responses of at most 512 tokens. Table 2 shows that its gains carry over. Allowing up to 8,192 tokens per response, ReFit-full raises the acceptance length from 1.97 to 2.68 on R1-Distill and from 1.65 to 2.73 on Nemotron-Nano under greedy decoding. Under temperature sampling, where verification accepts tokens stochastically, it raises the acceptance length from 1.96 to 2.58 and from 1.64 to 2.63. With a 2,048-token budget, where we also evaluate the dedicated drafter, ReFit-full closes 52% of its gap on R1-Distill.

6.3 Scaling and Cost

Figure 4a shows that ReFit improves steadily with data. With the production drafter, the gap closed by ReFit-full grows from 46% with 0.5k examples to 58% with 4k, 66% with 16k and 74% with 64k. Every increase in data helps, and the curve is still rising at 64k examples. The EAGLE authors’ family drafter starts from a better position and stays ahead at every budget (64% at 4k, 72% at 16k). ReFit-interface follows a parallel curve 12–15 points lower and thus captures most of the benefit of the full warm start with an eighth of the trainable parameters.

Figure 4b places these results in terms of compute. Even 0.45 GPU-hours of repair on 256 exam-

Method	R1-Distill-Llama-8B						Nemotron-Nano-8B					
	Train GPU-h	p_1 (%)	τ		Speedup		p_1 (%)	τ		Speedup		
			SPEED	MATH	bs 1	bs 8		SPEED	MATH	bs 1	bs 8	
Autoregressive	—	—	1.00	1.00	1.00×	1.00×	—	1.00	1.00	1.00×	1.00×	
Independent drafter (1B)	—	61.3	2.44	2.97	1.26×	1.11×	64.7	2.65	2.84	1.37×	1.24×	
Family drafter, reused	0	43.0	1.76	1.93	1.31×	1.07×	42.9	1.76	1.70	1.29×	1.13×	
ReFit-interface (ours)	11	62.6	2.41	2.77	1.74×	1.27×	61.4	2.42	2.72	1.69×	1.39×	
ReFit-full (ours)	11	65.0	2.54	2.97	1.83×	1.29×	62.9	2.49	2.91	1.80×	1.42×	
Dedicated drafter (reference)	~4.5k	70.5	2.85	3.84	2.05×	1.41×	not available					

Table 1: **Main results.** First-token acceptance p_1 , acceptance length τ and end-to-end speedup over autoregressive decoding (vLLM, A40, batch sizes 1 and 8). MATH is MATH-500 for R1-Distill and a 64-problem MATH sample for Nemotron-Nano. ReFit improves every acceptance metric over reuse with 95% confidence intervals excluding zero, and ReFit-full is the fastest method that does not require a dedicated drafter. Training cost includes generating the repair data; the dedicated drafter’s cost is estimated for the public EAGLE-3 recipe.

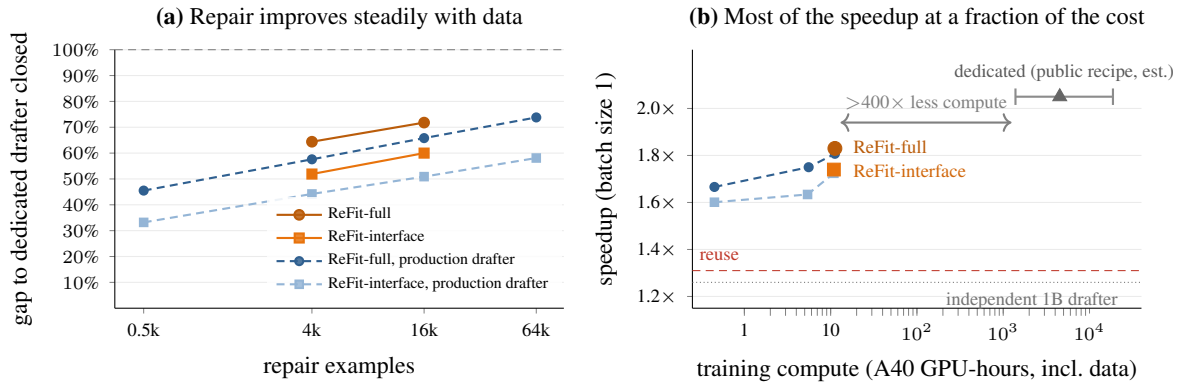


Figure 4: **(a)** Gap to the dedicated drafter closed on R1-Distill-Llama-8B as a function of the number of repair examples, for the EAGLE authors’ family drafter (solid) and the production drafter (dashed). **(b)** Batch-size-1 speedup against training compute, including data generation. ReFit attains most of the dedicated drafter’s speedup with orders of magnitude less compute; the dedicated drafter’s range spans 10 to 40 epochs of the public recipe.

Target	Decoding	Reuse	ReFit		Dedicated
			interface	full	
R1	greedy	1.97	2.59	2.68	3.78
	sampled	1.96	2.49	2.58	3.38
Nemotron	greedy	1.65	2.53	2.73	—
	sampled	1.64	2.42	2.63	—

Table 2: ReFit on long reasoning. Acceptance length on MATH problems with responses of up to 8,192 tokens (2.4k–3.3k on average), under greedy decoding and temperature sampling ($T=0.6$, top- p 0.95). ReFit is trained on responses of at most 512 tokens; every gain over reuse has a 95% confidence interval excluding zero.

ples already yield a $1.67\times$ speedup, and 11 GPU-hours reach $1.83\times$. By contrast, we estimate that reproducing the dedicated drafter with the public EAGLE-3 recipe requires between 1.3k and 19k A40 GPU-hours, depending on the number of epochs. ReFit thus recovers most of the dedi-

cated drafter’s speedup for two to three orders of magnitude less compute, and it can be applied the day a derivative is released.

6.4 Generality

Table 3 shows that the benefits of ReFit do not depend on a particular drafter. The production EAGLE-3 drafter distributed with vLLM breaks as much as the EAGLE authors’ drafter and is repaired almost as well: ReFit-full and ReFit-interface close 66% and 51% of the R1-Distill gap and reach speedups of $1.81\times$ and $1.72\times$. DFlash replaces autoregressive drafting with parallel block diffusion but reads the target through the same kind of feature interface, and repairing it with only 256 examples raises its acceptance length by 11–13% on both targets. ReFit also repairs drafters in a second model family: the Qwen3-8B family drafter, reused on DeepSeek-R1-0528-Qwen3-8B, gains 0.15 and 0.23 tokens per step from ReFit-interface

Family drafter	Target	Reuse	ReFit	
			interface	full
EAGLE-3 (authors)	R1	1.76	2.41	2.54
	Nemotron	1.76	2.42	2.49
EAGLE-3 (production)	R1	1.73	2.30	2.47
	Nemotron	1.82	2.32	2.42
DFlash (256 ex.)	R1	2.08	2.22	2.31
	Nemotron	2.08	2.24	2.36
EAGLE-3 (Qwen3)	R1-Qwen3	1.93	2.08	2.16

Table 3: ReFit generalizes across drafter checkpoints and drafter families. Acceptance length on SPEED-Bench; Llama-family EAGLE-3 drafters use 16k repair examples; DFlash, which drafts blocks of ten tokens, and the Qwen3-8B EAGLE-3 drafter on DeepSeek-R1-0528-Qwen3-8B are repaired with only 256 examples.

and ReFit-full with only 256 examples. Across all seven settings, the pattern of Figure 2 holds: the interface carries most of the repair, and a full warm start adds the rest.

7 Related Work

Speculative decoding. Speculative decoding (Leviathan et al., 2023; Chen et al., 2023) has been extended with tree-structured verification (Miao et al., 2024; Chen et al., 2024), self-speculation (Zhang et al., 2024a), and model-free drafting from n-grams, retrieval or suffix structures (Fu et al., 2024; He et al., 2024; Saxena, 2023; Oliaro et al., 2024); see Xia et al. (2024) for a survey. The strongest drafters condition on the target’s hidden states, from Medusa and Hydra heads (Cai et al., 2024; Ankner et al., 2024) to the EAGLE family (Li et al., 2024b,a, 2025; EAGLE Team et al., 2026), harmonized representations (Zhang et al., 2024b), parallel drafters (Jiang et al., 2026) and block-diffusion drafters (Chen et al., 2026b). Open frameworks make training such drafters possible but not cheap (Li et al., 2026a; Red Hat AI and vLLM Project, 2025), and pruning the draft vocabulary reduces their cost (Zhao et al., 2025; Goel et al., 2025). We take these drafters as given and keep them working when their target changes.

Adapting drafters. DistillSpec (Zhou et al., 2024) distills a draft model from its target, online speculative decoding (Liu et al., 2024) keeps updating it on served traffic, and Draft-OPD (Lei et al., 2026) trains drafters on their own on-policy proposals. Other work aligns draft models with chat-tuned targets (Goel et al., 2024), trains drafters for new domains (Hong et al., 2025), adapts them with

parameter- and data-efficient updates (Lin et al., 2026), shares frozen drafts across evolving targets (Li et al., 2026b), or evolves them online across vocabularies and targets (Ramakrishnan et al., 2025; Zhang et al., 2026). These approaches re-align the drafter as a whole or change its architecture or training regime. For target-conditioned drafters, we identify a single existing component, the feature interface, as the most effective place to repair, and show that re-fitting it offline from public instructions recovers most of a dedicated drafter’s speedup without traffic logs, new modules or original training data.

Post-training and representation drift. Fine-tuning distorts pretrained features (Kumar et al., 2022), and the extent depends on the training signal: on-policy RL forgets less than supervised training on fixed data (Shenfeld et al., 2026; Chen et al., 2026a). The open-model ecosystem consists largely of such derivatives (Horwitz et al., 2025), and reasoning distills (DeepSeek-AI, 2025; Bercovich et al., 2025) and multi-stage post-training pipelines (Lambert et al., 2024) are among its most widely deployed members. Our census connects these findings to inference efficiency. The same regimes that move a model furthest from its initialization are the ones that break the drafters built for it.

8 Conclusion

Family drafters make speculative decoding practical across the open-model ecosystem, but they silently lose much of their value on derivatives distilled from stronger reasoning teachers. We showed that post-training shifts both the features a drafter reads and the tokens it must predict, that the drafter’s feature interface is the most effective place to repair it, and that re-fitting this interface on the derivative’s own responses to public instructions restores most of a dedicated drafter’s speedup for a few GPU-hours. ReFit turns per-derivative drafting from an expensive training project into a routine step that can accompany every model release. More broadly, our results suggest that the coupling that makes target-conditioned drafters fast need not make them brittle: when the target moves, it suffices to teach the drafter how to read it again.

Limitations

Our study focuses on 8B-scale derivatives of the Llama-3.1 and Qwen3 families and on the EAGLE-

3 and DFlash drafters, which are among the most widely deployed configurations. ReFit requires running the derivative to generate responses; this cost is small and is included in all our compute figures. We measure wall-clock speedups with greedy decoding on NVIDIA A40 GPUs in vLLM; absolute speedups vary with hardware, batch size and decoding configuration.

Ethical Considerations

ReFit leaves the target model’s weights and outputs unchanged: speculative decoding is lossless, so repair affects only serving cost, not what the model generates. By reducing the compute needed to serve reasoning models, it lowers the energy cost of deploying them. All models and prompt sources are public and used under their licenses, and no private user data is involved.

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A Implementation Details

Repair. We optimize Eq. (2) with AdamW (learning rate 2×10^{-5} , weight decay 0.01, cosine schedule with 3% warm-up) for one epoch over the 16k repair examples, packing roughly 2k response tokens per step, with $k_{\max}=3$ training-time-test steps. The loss is computed on response positions only. Repair data are generated greedily with vLLM, capped at 512 response tokens, and deduplicated exactly and approximately against all evaluation prompts, including the full MATH-500 set. ReFit-interface updates $W_{fc} \in \mathbb{R}^{4096 \times 12288}$ (50.3M parameters); ReFit-full updates the interface, the decoder layer and the language-modeling head (424.8M parameters), keeping the token embedding shared with the target. When a released drafter omits the embedding table, the trainer loads the target’s embedding, exactly as the serving engine does. The three R1-Distill seeds use the same data and differ only in the training seed.

Component analysis. For Figure 2, all variants use the same 256 R1-Distill responses, batches, learning rate and 300 optimization steps, starting from the production EAGLE-3 drafter. Interface variants are LoRA updates of W_{fc} with ranks 8, 32 and 75 (0.13M, 0.52M and 1,228,800 parameters) and the dense interface (50,331,648). Decoder

variants are LoRA on the query, value and MLP projections with rank 16 (1,228,800 parameters, exactly matching the rank-75 interface) and rank 655 (50.30M), a full-rank update of the query and output projections (50,331,648, exactly matching the dense interface), and the whole decoder layer (243.3M); the interface and head stay frozen in all decoder variants. The feature-statistics baseline fits a per-layer affine map that matches the mean and scale of the derivative’s tapped features to the parent’s. The parent-supervision control keeps the derivative-generated token sequences, masks, batches, initialization and objective fixed and replaces only the model that supplies the hidden states and next-token distributions.

Crossover. For each model, we take 64 contexts generated by the derivative, run both the parent and the derivative on the identical text, and feed each model’s tapped hidden states to the frozen family drafter. Agreement is the rate at which the drafter’s first prediction matches a given model’s greedy next token. The feature-source effect averages the change in agreement from swapping parent features for derivative features under each fixed verifier; the policy effect averages the change from swapping the verifier under each fixed feature source.

Long reasoning. Table 2 uses the first 32 problems of the MATH sample with response budgets of 512, 2,048 and 8,192 tokens, evaluated with the same engine, prompts and drafters as the main results. Sampled decoding uses temperature 0.6 and top- p 0.95 with a fixed seed. No response in any arm exceeds 50% repeated 4-gram coverage.

Training-time-test depth. The drafter is unrolled for $k_{\max}=3$ steps during training and proposes $K=4$ tokens at serving time. Training with $k_{\max}=4$ instead gives the same acceptance ($\Delta\tau = -0.004$ and -0.007 on SPEED-Bench for ReFit-interface and ReFit-full at 4k examples, both within noise).

Evaluation. All acceptance measurements use vLLM with greedy decoding, $K=4$ draft tokens (DFlash: its native block of ten), a maximum of 512 new tokens and batch size 8. Prompts that produce no speculative step are excluded pairwise across compared methods. Speedups are measured on 32 SPEED-Bench prompts at batch size 1 and 128 at batch size 8, each in three fresh server processes with three warm repetitions, relative to autoregressive decoding of the same prompts; confidence

intervals resample processes and prompt batches jointly. The dedicated drafter’s training cost is estimated from our measured A40 training throughput for the public EAGLE-3 recipe at 10 to 40 epochs.

B Census Details

The census covers 174 derivatives, 87 in each of the Llama-3.1-8B and Qwen3-8B families, that load with the architecture and tokenizer of the instruction-tuned parent. Post-training histories are taken from model cards and grouped as in Figure 1a; models whose cards describe several stages are grouped by their combined history, and models without a documented history are excluded from the grouped analysis. Retention is computed per derivative on 64 prompts from its own domain and on 128 SPEED-Bench prompts, and aggregated within each group. The two reasoning distills are evaluated on SPEED-Bench with 128 prompts. The compatibility probe ranks derivatives by acceptance retention on 16 fixed requests; we report its AUROC for identifying derivatives whose retention on the remaining, disjoint requests falls below 90%.